IB MATHEMATICS SL • UNIT 3.01

C Topic 3: Geometry, Trigonometry & Voronoi Diagrams

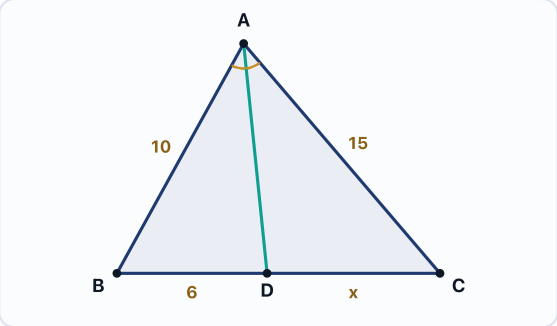
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Topic 3: Geometry, Trigonometry & Voronoi Diagrams Practice Exercises

1 MCQ

In the given triangle ABC, AB = 10, AC = 15, and BD = 6. D is a point on BC such that AD is the angle bisector of angle BAC. Find the length of DC.



A 9

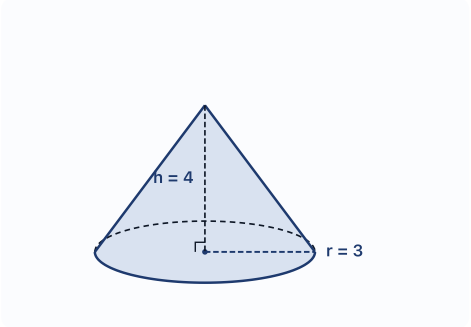
B 10

C 15

D 22.5

2 MCQ

Volume of cone with radius 3, height 4:



A 12π

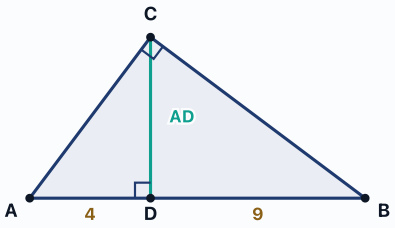
B 16π

C 36π

D 48π

3 MCQ

In the given right-angled triangle ABC, angle B is 90 degrees. Point D is on BC such that BD = 4 and DC = 9. If AD is the altitude to the hypotenuse AC, find the length of AD.



A 8

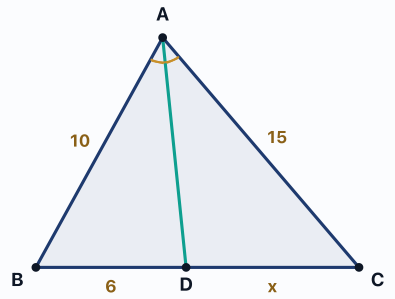
B 12

C 6

D 15

4 MCQ

In the triangle ABC, AB = 10, AC = 15, BD = 6, and D is a point on BC such that AD is the angle bisector of angle BAC. Find the length of DC.



A 12

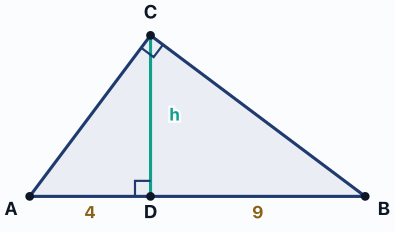
B 7

C 9

D 10

5 MCQ

In the diagram, triangle ABC has angle ABC equal to 90 degrees. Point D lies on AC such that BD is perpendicular to AC. If AD = 4 cm and DC = 9 cm, find the length of BD.



A 6 cm

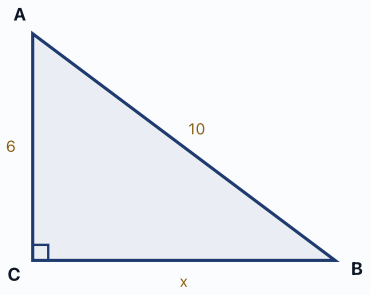
B 5 cm

C 7 cm

D 3 cm

6 MCQ

In the right-angled triangle ABC shown below, the length of side AB is 10 units and the length of side BC is 6 units. The angle ACB is 90 degrees. Find the length of side AC.



A 10

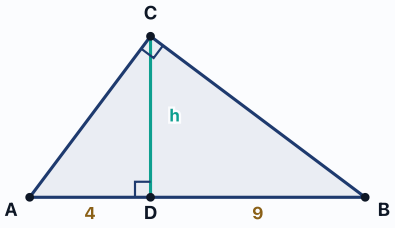
B 6

C 16

D 8

7 MCQ

In the right-angled triangle ABC, shown below, AD is the altitude to the hypotenuse BC. If BD = 4 and DC = 9, find the length of the altitude AD.



A 6

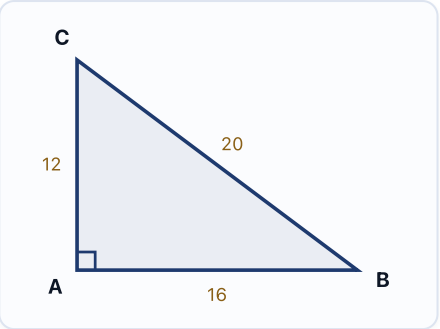
B 8

C 5

D 7

8 MCQ

A right triangle has legs 12 and 16 and hypotenuse 20. The angle θ is opposite the side of length 12. Find $\cos(\theta)$.



A $\frac{4}{3}$

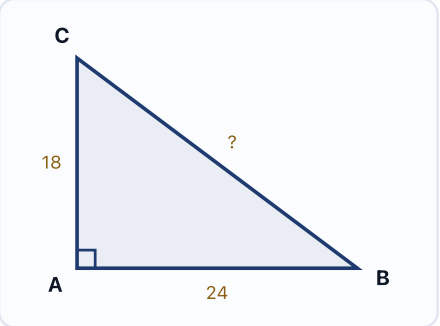
B $\frac{3}{5}$

C $\frac{4}{5}$

D $\frac{3}{4}$

9 MCQ

A right triangle has legs of length 18 and 24. Find the length of the hypotenuse.



A 34

B 33

C 30

D 32

10 MCQ

Convert 210° to radians, giving your answer as an exact multiple of π.

A $\frac{7\pi}{6}$

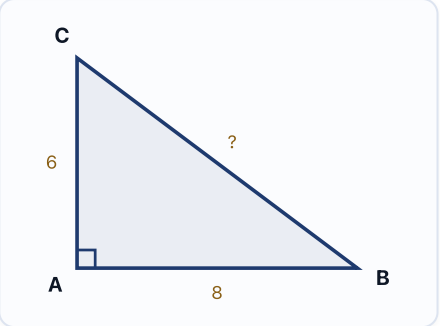
B $\frac{5\pi}{3}$

C $\frac{\pi}{2}$

D $\frac{7\pi}{4}$

11 MCQ

A right triangle has legs of length 6 and 8. Find the length of the hypotenuse.



A 5

B 10

C 7

D 0

12 MCQ

Convert 270° to radians, giving your answer as an exact multiple of π.

A $\frac{3\pi}{4}$

B $\frac{\pi}{4}$

C $\frac{3\pi}{2}$

D $\frac{2\pi}{3}$

13 MCQ

Convert $\frac{\pi}{2}$ radians to degrees.

A 30°

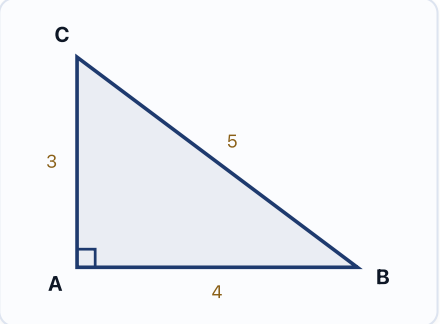
B 240°

C 90°

D 315°

14 MCQ

A right triangle has legs 3 and 4 and hypotenuse 5. The angle θ is opposite the side of length 3. Find $\cos(\theta)$.



A $\frac{3}{4}$

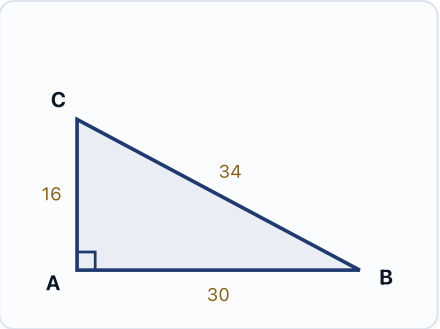
B $\frac{3}{5}$

C $\frac{4}{3}$

D $\frac{4}{5}$

15 MCQ

A right triangle has legs 16 and 30 and hypotenuse 34. The angle θ is opposite the side of length 16. Find $\sin(\theta)$.



A $\frac{8}{15}$

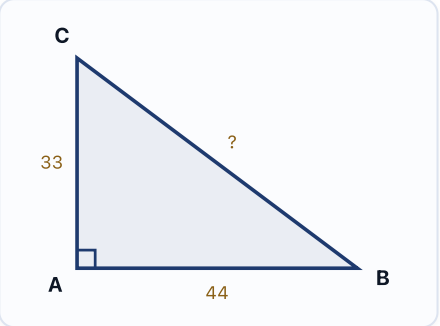
B $\frac{15}{8}$

C $\frac{15}{17}$

D $\frac{8}{17}$

16 MCQ

In an applied problem, a right triangle has legs of length 33 and 44. Find the length of the hypotenuse.



A 50

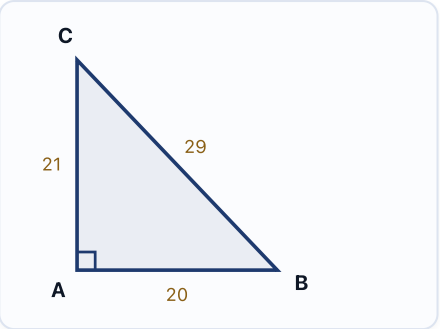
B 45

C 65

D 55

17 MCQ

A right triangle has legs 21 and 20 and hypotenuse 29. The angle θ is opposite the side of length 21. Find $\sin(\theta)$.



A $\frac{21}{20}$

B $\frac{20}{21}$

C $\frac{20}{29}$

D $\frac{21}{29}$

18 MCQ

A right triangle has one leg of length 12 and hypotenuse 20. Find the length of the other leg.

A 13

B 6

C 11

D 16

19 MCQ

A right triangle has one leg of length 9 and hypotenuse 15. Find the length of the other leg.

A 12

B 8

C 7

D 2

20 MCQ

Convert 240° to radians, giving your answer as an exact multiple of π.

A $\frac{7\pi}{6}$

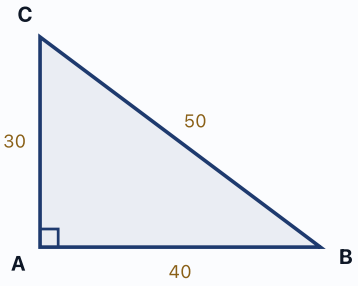
B π

C $\frac{4\pi}{3}$

D $\frac{\pi}{3}$

21 MCQ

In an applied problem, a right triangle has legs 30 and 40 and hypotenuse 50. The angle θ is opposite the side of length 30. Find $\cos(\theta)$.



A $\frac{4}{3}$

B $\frac{4}{5}$

C $\frac{3}{4}$

D $\frac{3}{5}$

22 MCQ

In an applied problem, a right triangle has one leg of length 9 and hypotenuse 41. Find the length of the other leg.

A 36

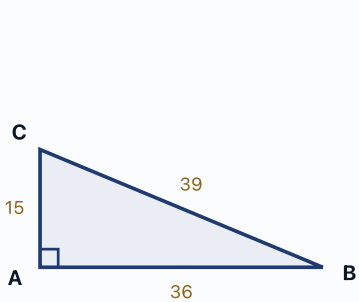
B 35

C 37

D 40

23 MCQ

A right triangle has legs 15 and 36 and hypotenuse 39. The angle θ is opposite the side of length 15. Find $\tan(\theta)$.



A $\frac{15}{38}$

B $\frac{5}{12}$

C $\frac{17}{36}$

D $\frac{12}{5}$

24 MCQ

Convert 330° to radians, giving your answer as an exact multiple of π.

A $\frac{11\pi}{6}$

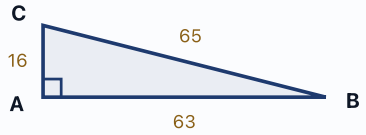
B $\frac{\pi}{2}$

C $\frac{4\pi}{3}$

D $\frac{\pi}{3}$

25 MCQ

A right triangle has legs 16 and 63 and hypotenuse 65. The angle θ is opposite the side of length 16. Find $\tan(\theta)$.



A $\frac{16}{63}$

B $\frac{2}{7}$

C $\frac{63}{16}$

D $\frac{16}{65}$